Istanbul Medipol University

School of Engineering and Natural Sciences

Communication Systems – Spring 2025
Project Final Report

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Video link:
https://drive.google.com/file/d/1UFftarSBQ4mKz1szlB2qc4DZ3nEjhd9P/view?usp=sharing

1. Introduction

In digital communication systems, Binary Phase Shift Keying (BPSK) is one of the most fundamental and widely used modulation schemes due to its simplicity. BPSK transmits data by shifting the phase of a carrier signal by 180 degrees, representing binary 0 and 1 with distinct phase states.

A key performance metric in digital communication is the Bit Error Rate (BER), which indicates the likelihood of a transmitted bit being incorrectly decoded at the receiver. BER is affected by many factors, including noise power, signal strength, and channel conditions. Additive White Gaussian Noise (AWGN) is one of the most common noise models used to represent the noise in real-world scenarios.

The project focuses on simulating the performance of a BPSK system over an AWGN channel on MATLAB and verifying the results using a Software Defined Radio (SDR) platform. The goal is to compare theoretical BER results with those measured through real-time transmission and reception.

This simulation shows how changes in noise power affect the BER and examines additional performance metrics including SNR calculation, signal and noise power, constellation plots, and bit decisions. Furthermore, the system's behavior under wireless transmission is evaluated to determine how real-world channel impairments influence performance.

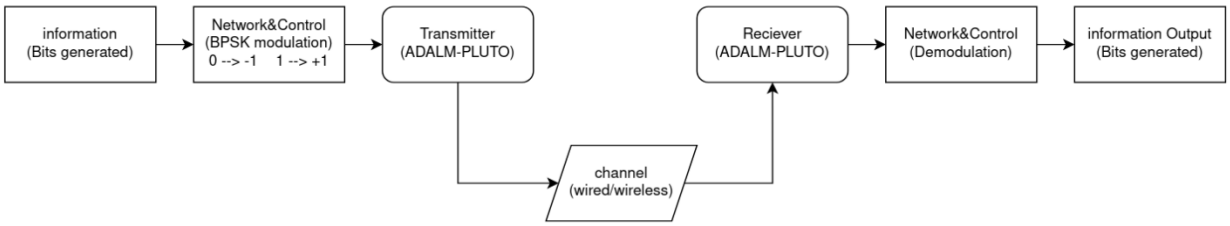

Figure 1: The Digital Communication Systems

The communication process begins with creating a stream of binary bits (0s and 1s). This data is then passed through the Network & Control block, where BPSK modulation is applied: the binary '0' is mapped to -1 and '1' to +1. These modulated symbols are transmitted via the ADALM-PLUTO which transmits the signal through a channel. On the receiving side, the ADALM-PLUTO receiver captures the transmitted signal and sent it to another Network & Control block, which performs demodulation, converting the signal back into binary form. Finally, the information output block reproduces the original bitstream, completing the communication process.

2. Blocks and Their Functionality

A complete BPSK (Binary Phase Shift Keying) simulation was completed in MATLAB to analyze the performance under AWGN (additive white Gaussian noise). Each block in the simulation corresponds to a functional stage of the digital transmission system. Key parameters are defined, and their impact on system performance is observed by varying them.

The bit generation block creates a random sequence of N = 1e5 bits. A larger N improves BER accuracy, while a smaller value may lead to unstable results. The BPSK modulator maps bits to symbols using symbols = 2*bits - 1, where 0 → –1 and 1 → +1.

The AWGN channel adds noise to simulate real-world transmission conditions. The noise level is controlled via the Eb/N0 parameter, and varying it allows to study BER under different SNR scenarios. Higher Eb/N0 values produce less noise and lower BER, while lower values increase the error rate.

At the receiver, a simple threshold detector compares received values to zero (decoded_bits = received > 0). The BER calculator then compares transmitted and received bits to determine performance.

Additional blocks computed the average power of the transmitted signal, noise, and received signal to calculate the actual SNR, which is compared to theoretical values. A constellation plot shows the distribution of received symbols; widespread at low Eb/N0 indicates more errors, while clustering at high Eb/N0 indicates reliable reception.

After finishing the simulation, it was time for the implementation which was done on gnu-radio companion which provides easy and intuitive python-based block integration. Below the system block diagram will be shown (Figure 2) and stating the purpose and function of each block.

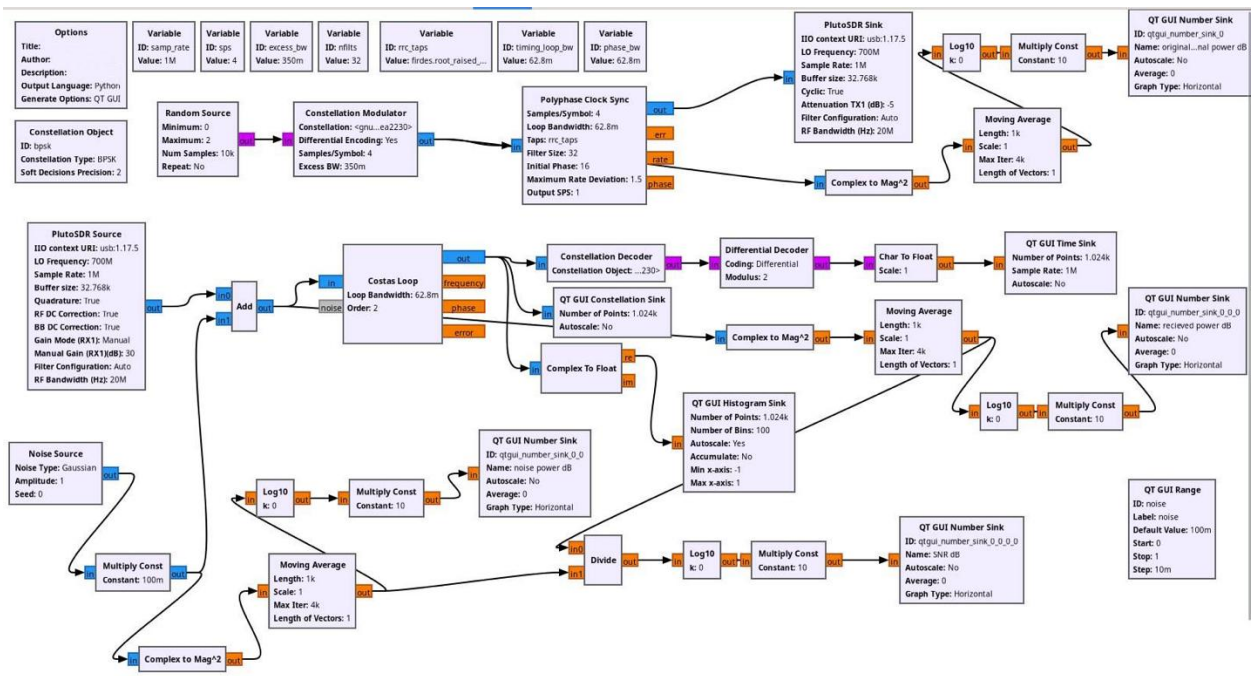

Figure 2: block diagram of the system using gnu-radio companion.

- **Random source:** generates random stream of bit 0 1, this represents the information in communication systems.
- **Constellation modulator:** this block maps 0 –> -1 & 1 –> +1, which represents the digital modulator for the system.
- **Polyphase Clock Sync:** this block performs symbol timing recovery, to align the samples with the symbol rate
- **Pluto SDR Sink/Source:** this block is the interface to the adalm-pluto, this represents the channel in typical communication systems
- **Noise source:** since we want to have control over noise, this block simulates AWGN which is present in everyday applications.
- **Costas Loop:** this block tracks and corrects frequency/phase offsets introduced from channel. This block is essential for synchronization and often implemented as PLLs in hardware.
- **Constellation Decoder:** determines the closest valid symbol by deciding the waveform received, which represents the demodulator.
- **Differential Decoder:** this block is critical because devices such as pluto SDR don't know the absolute phase so the received bits could be 180 degrees shifted(flipped) and so by using differential encoding/decoding we can eliminate that possibility and always get the correct bits

This system uses 4samples/symbol, this oversampling provides robustness for the overall system and protects the system from phase/frequency offset, the carrier frequency used is 700MHz as shown in the SDR configuration in Figure 2.

The figure below shows a running simulation where we can control the noise magnitude and see the effects on the histogram constellation and scatter plot along in real time, along with signal in time domain.
The figure also shows the original signal power, noise power, and calculated SNR after introducing the AWGN to the received signal.

3. Simulation/Demo Results

3.1 Matlab Simulations:

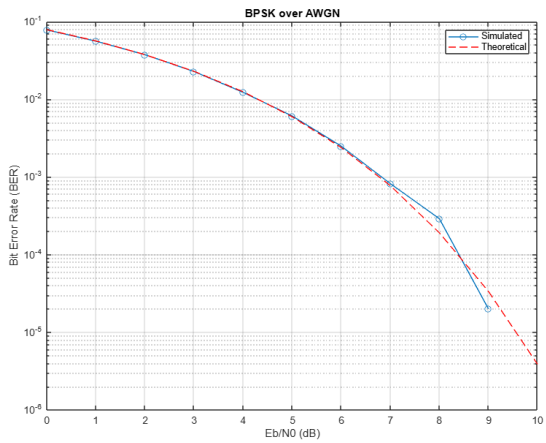

Figure 3: The plot is a Bit Error Rate (BER) vs Eb/N0. It shows that as Eb/N0 increases, the BER decreases exponentially.

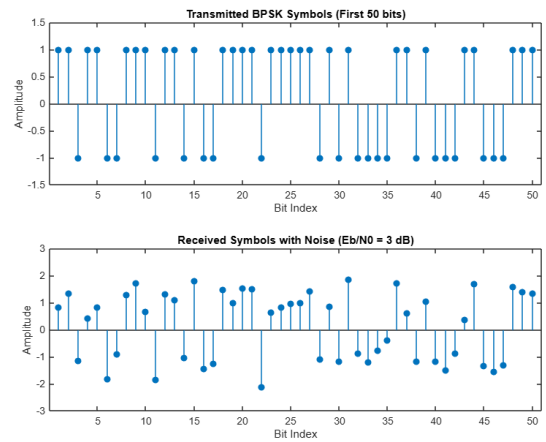

Figure 4: The subplot compares the first 50 transmitted BPSK symbols (±1) with the corresponding noisy received samples. It visually demonstrates how AWGN distorts the signal amplitude, especially at lower Eb/N0.

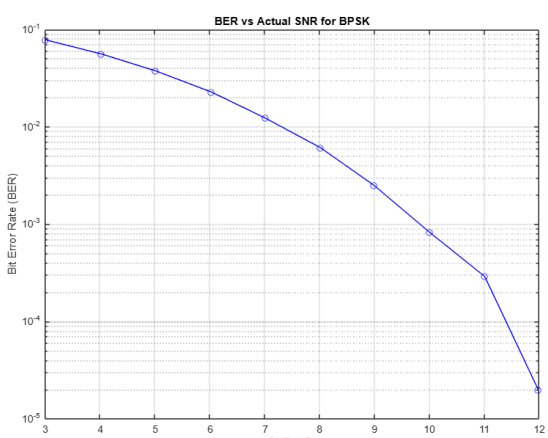

Figure 5: The plot shows the BER versus the actual measured SNR, calculated from signal and noise powers.

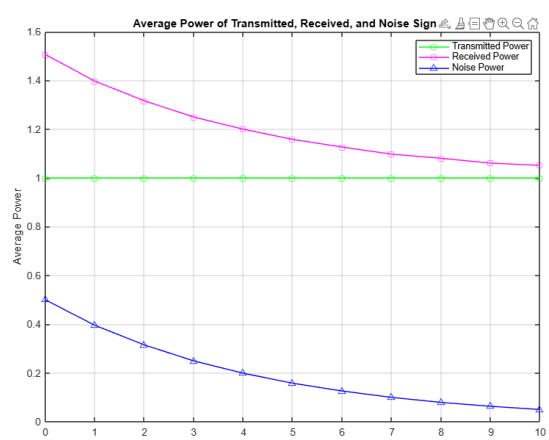

Figure 6: The plot shows how the transmitted power (P_{tx}), received power (P_{rx}), and noise power (P_{noise}) change with Eb/N0. It confirms that transmitted power remains constant, while received and noise power decreases with increasing Eb/N0.

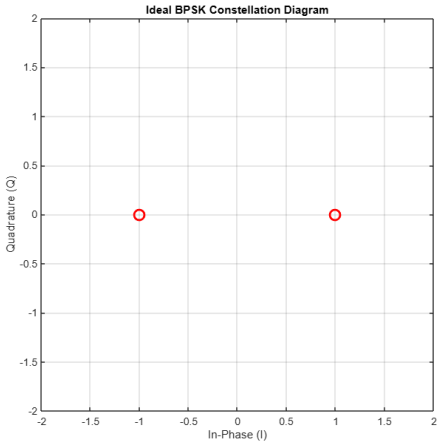

Figure 7: The plot shows the theoretical BPSK constellation with two ideal points at -1 and +1 on the real axis.

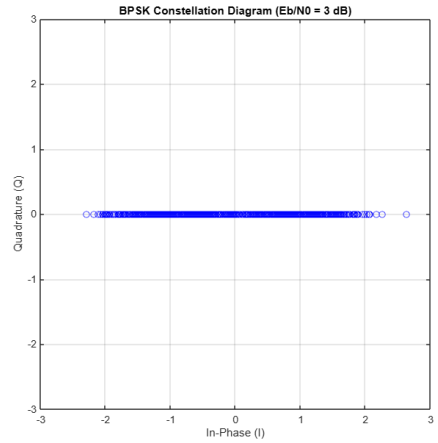

Figure 8: The plot shows a noisy constellation for BPSK at a mid-level Eb/N0 (3 dB). The symbols received are scattered around the ideal points due to AWGN. This visualizes how noise affects symbol decision boundaries and BER performance.

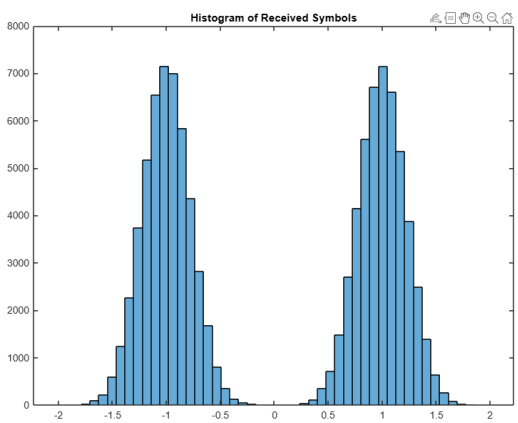

Figure 9: The histogram shows the distribution of the received symbols. It illustrates the Gaussian nature of noise centered around the two symbol values (-1 and +1).

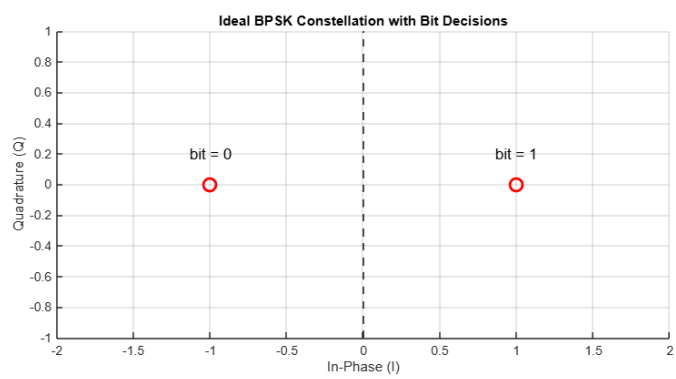

Figure 10: The plot shows the ideal BPSK constellation along with the decision boundary at 0 on the real axis since it's the mid point.

3.2 Demo Results

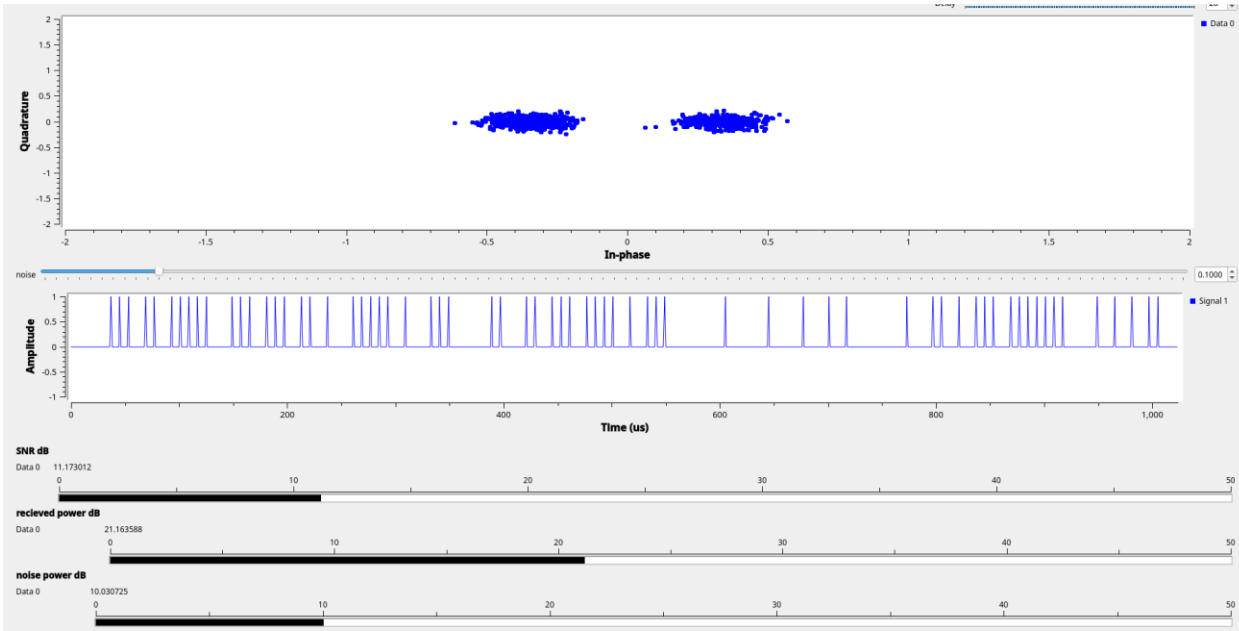

Figure 11: this figure shows the constellation cluster after bit decision with its relative SNR, signal power and Noise power respectively, as well as signal received in time domain.

In the above figure the SNR is about 11 and you can see a clear distinction between a bit of decision clusters.

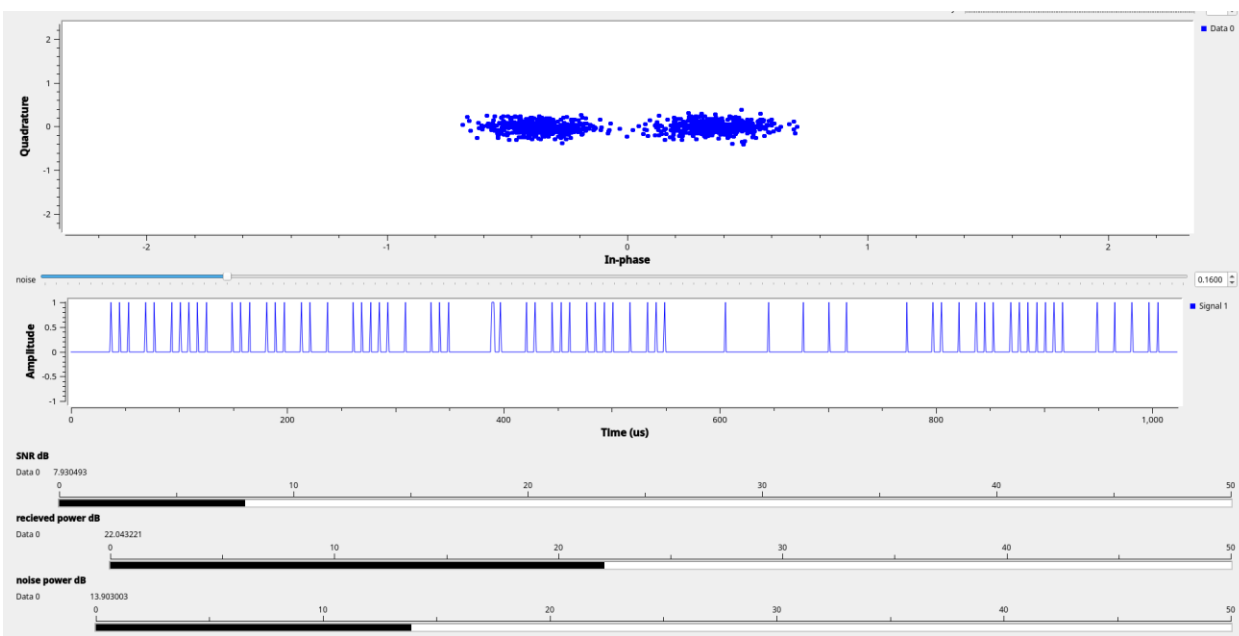

Figure 12: When the noise increased.

As we increase the noise and thereby decrease the SNR the clusters come closer together, decreasing the SNR any further results in higher bit error rate where the decision is not clear between the bits.

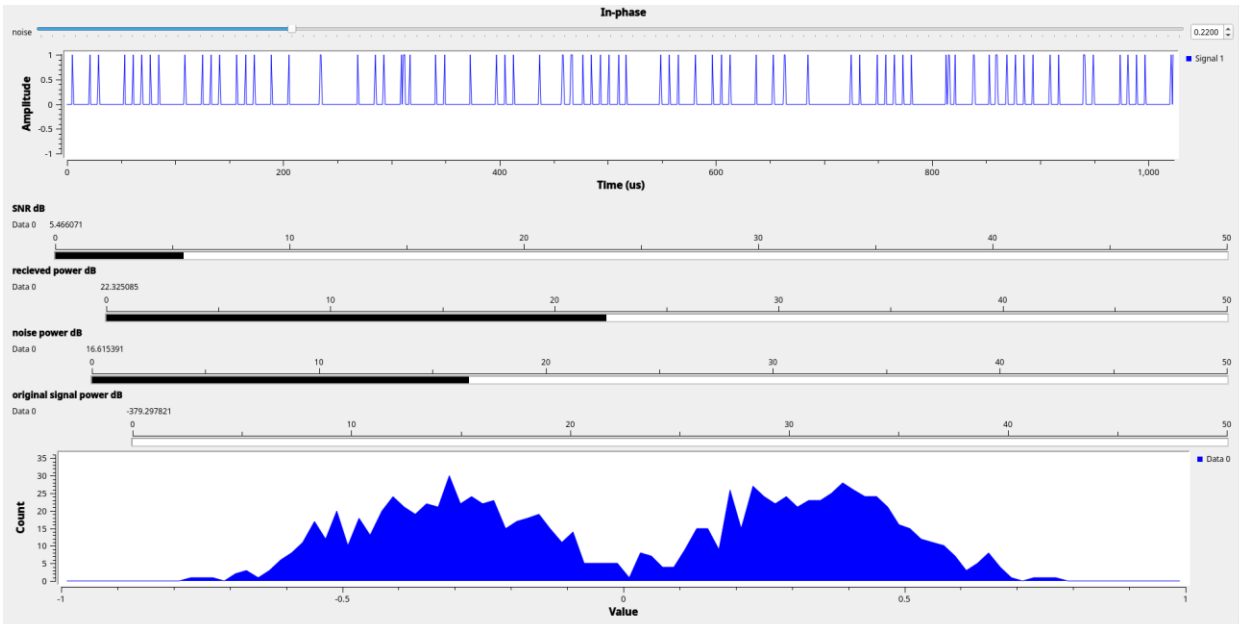

Figure 13: different visualization showing histogram.

The more we decrease the SNR the two curves move closer together resulting in higher BER.

Difficulties faced: during the making of the gnu diagram there's a specific block called BER which is supposed to calculate the bit error rate, however adding this block causes the graphical results to halt and always display 0, despite the really low SNR, unfortunately looking at the documentation of the block didn't help as it's only used for specific applications and isn't supported very well.
In order to overcome this issue, we tried to run the SDR and calculate the BER straight from python code, however the issue wasn't fixed.

The BPSK simulation worked well in MATLAB, but the results slightly differed when using the ADALM-PLUTO SDR due to real-world imperfections. MATLAB provides an ideal environment, while SDR introduces noise, frequency offsets, and hardware non-idealities. This led to **non-ideal constellation plots** in the SDR setup. Despite these challenges, the system functioned correctly, showing the practical impact of real-world effects on theoretical models.

Conclusion

In conclusion This project successfully designed, implemented, and validated a complete BPSK communication system using (SDR) technology. The system uses the ADALM Pluto SDR platform and GNU Radio Companion (GRC) software to demonstrate fundamental principles of digital communication systems. The project validates characteristic BER/SNR relationship observed in both MATLAB and SDR implementation; the system demonstrates operations such as encoding, pulse shaping and synchronization. In order to make the design better, better channel encoding can be used and extending the system to QBSK plus incorporating equalizers for multi-path mitigation.